

BHAVJEET SINGH

my-portfolio-pink-pi-65.vercel.app | bhavjeetsingh784@gmail.com | +91 7678205626 | Delhi

PROFESSIONAL SUMMARY

AI/ML Backend Engineer pursuing B.Sc. Data Science at IIT Madras, specializing in building and deploying LLM applications, RAG pipelines, and ML systems on AWS and Azure. Fine-tuned a local 3B model to **outperform GPT-4o-Mini**, **cut legal contract review time by 60%**, and built a demand forecasting pipeline saving **INR 62 crore/hour** in grid penalties.

EDUCATION

Indian Institute of Technology, Madras <i>Bachelor of Science in Data Science and Applications</i>	2025 – Present
--	----------------

PROJECTS

ToolCaller-India – Local Function-Calling LLM | Python, PyTorch, LoRA/PEFT, Hugging Face, Gradio

- Fine-tuned a 3B-param Qwen model with LoRA on a curated 13.3k-example SFT dataset (8.8k custom multi-turn Indian API calls – UPI, IRCTC, Aadhaar) to handle low-latency function calling locally.
- Hit **70.4% weighted accuracy** on the BFCL dev split with **100% valid JSON output**, beating GPT-4o-Mini (69.1%) at a fraction of the inference cost.
- Built a live Gradio dashboard for interactive testing; open-sourced the full training pipeline and evaluation harness on GitHub.

ContractIQ – Legal AI Data Pipeline Service | Python, FastAPI, LangChain, FAISS, Docker, AWS ECS/Fargate

- Built a RAG pipeline with FastAPI, LangChain, and FAISS that ingests and indexes 500+ unstructured legal PDFs for clause extraction and risk flagging.
- Sped up LLM inference **by 40%** through vLLM quantization and Redis response caching; wired LangSmith for per-request token and latency tracking.
- Reduced average contract review turnaround **from hours to minutes (60% faster)**, holding P50 latency at 2.8s and cost at **\$0.08 per 67k tokens** via Gemini routing.
- Deployed on AWS ECS Fargate with CloudFormation IaC and GitHub Actions CI/CD – **zero-downtime rolling updates** in production.

24-Hour Electricity Demand Prediction – ML Pipeline | Python, XGBoost, Prophet, LSTM, MLflow, FastAPI, Azure

- Forecasted India’s 180 GW grid demand 24 hours ahead at **6.91% MAPE** – each 1% improvement saves roughly **INR 62 crore/hour** in settlement penalties.
- Engineered 19k+ hourly features (lag windows, weather joins, rolling stats) from raw IEX and Open-Meteo data using PySpark batch transforms.
- Benchmarked XGBoost, Prophet, and LSTM across **14+ MLflow-tracked experiments** with full artifact versioning for reproducibility.
- Shipped an automated retrain-and-deploy pipeline via GitHub Actions to Azure, serving predictions through a FastAPI endpoint.

TECHNICAL SKILLS

Languages: Python, SQL, Nextjs, Bash, Git
ML/DL: PyTorch, Scikit-Learn, XGBoost, Hugging Face, LoRA/PEFT, MLflow
GenAI: LangChain, LangGraph, CrewAI, RAG, AI Agents, Function Calling, Prompt Engineering, vLLM
APIs & Vector DBs: OpenAI, Gemini, Groq, Anthropic, FAISS, Pinecone, Qdrant, ChromaDB
Backend: FastAPI, Flask, REST APIs, Pydantic, Redis, PostgreSQL
Cloud & DevOps: AWS (ECS, Fargate, EC2, S3), Azure, Docker, GitHub Actions, CI/CD, LangSmith

CERTIFICATIONS

LLMOps Live Cohort Program	December 2025 – April 2026
Ultimate Live RAG Bootcamp: Agentic Systems with Cloud Deployment	Jan 2026 – April 2026